

Computer Networks

CS360

Project-Phase 2

Deadline: Day 09/08/2026 @ 23:59

[Total Mark is 15]

Student Details:

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g., misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instruction

Project Objective:

Design, configure, and test a network setup in **Packet Tracer** to successfully transmit data packets between devices in different LANs across two separate WANs (WAN 1 and WAN 2).

Group Details

- Group Size = Maximum of 3 Members.
- One group member (group leader/coordinator) must **submit the project report** on blackboard. Marks will be given based on your submission and quality of the contents.

Project Report

- The project is divided into two phases:
 - Phase 1 (Mid project Progress Report)
 - Phase 2 (Final project Report)
- This report is for Phase 2. In this report, you should complete all the requirements for Phase 2.
- Each Report will be evaluated according to the marking criteria

*Learning
Outcome(s):*

*Demonstrate
protocol
configuration,
network-
addressing
schemes and
analyze packet
transmission for
the Local and
Wide Area
Networks.*

Project Description (Common to Phase 1 and Phase 2)

A university operates two campuses: a Main Campus and a Branch Campus. Each campus network is treated as a separate WAN site. Your task is to design and configure, in Cisco Packet Tracer, a network that successfully transmits data packets from a device in any LAN of WAN 1 (Main Campus) to a device in any LAN of WAN 2 (Branch Campus). Each WAN contains 2 LANs, and each LAN has 3 end devices.

Network Design Requirements:

1. WAN 1 (Main Campus):

- Contains LAN 1 (Computer Lab) and LAN 2 (Admin Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.10.0/24 for LAN 1 and 172.16.20.0/24 for LAN 2).
- Connect each LAN to a router (Router 1).

2. WAN 2 (Branch Campus):

- Contains LAN 3 (Library) and LAN 4 (Faculty Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.30.0/24 for LAN 3 and 172.16.40.0/24 for LAN 4).
- Connect each LAN to a router (Router 2).

3. WAN Connection:

- Connect Router 1 and Router 2 using a serial link to simulate a WAN connection.
- Assign IP addresses to the serial interfaces (e.g., 10.10.10.1/30 for Router 1 and 10.10.10.2/30 for Router 2).

Configuration Details:

1. Assign IP Addresses:

- Configure IP addresses, subnet masks, and default gateways for all devices in each LAN. Example:
 - LAN 1: 172.16.10.0/24, Default Gateway: 172.16.10.1 (Router 1's interface).
 - LAN 2: 172.16.20.0/24, Default Gateway: 172.16.20.1 (Router 1's interface).

- LAN 3: 172.16.30.0/24, Default Gateway: 172.16.30.1 (Router 2's interface).
- LAN 4: 172.16.40.0/24, Default Gateway: 172.16.40.1 (Router 2's interface).

2. Router Configuration:

- Configure IP addresses on the LAN and WAN interfaces of both routers. Example:
 - Router 1:
 - LAN 1 Interface: 172.16.10.1/24
 - LAN 2 Interface: 172.16.20.1/24
 - WAN Interface (Serial): 10.10.10.1/30
 - Router 2:
 - LAN 3 Interface: 172.16.30.1/24
 - LAN 4 Interface: 172.16.40.1/24
 - WAN Interface (Serial): 10.10.10.2/30

3. Enable Routing:

- Configure static routes or a dynamic routing protocol (e.g., RIP) on both routers to ensure communication between WAN 1 and WAN 2.

Phase 2 Requirements

(15 Marks)

Testing and Verification

1. Use the ping command to test connectivity between a device in any LAN of WAN 1 and a device in any LAN of WAN 2.
2. Verify that data packets are successfully transmitted (0% packet loss).

Questions to Answer (10 Marks — 2 marks each)

Answer all five questions. For each question, you must include (a) a clear screenshot from your Packet Tracer project and (b) the commands used at that instance (e.g., IOS CLI commands or the configuration steps applied). Screenshots without the corresponding commands/steps will receive ZERO marks for that question.

1. **Creating nodes:** Show a screenshot of the complete topology with all nodes created (12 end devices, 4 switches, 2 routers). List the device types selected and any steps/commands used to name the devices (e.g., hostname configuration). (2 marks)
 2. **Giving connections in the LAN:** Show a screenshot of the LAN connections between end devices, switches, and routers. State the cable types used for each connection and why. (2 marks)
 3. **Assigning IP addresses:** Show screenshots of IP address, subnet mask, and default gateway assignment on at least one end device in each LAN, and the commands used to configure the router LAN interfaces (e.g., interface, ip address, no shutdown). (2 marks)
 4. **Creating the WAN connection:** Show a screenshot of the serial link between Router 1 and Router 2, and the commands used to configure the serial interfaces and enable routing (static routes or a dynamic routing protocol such as RIP), including clock rate if applicable. (2 marks)
 5. **Successful data transmission using the ping command:** Show screenshots of successful ping tests between a device in WAN 1 and a device in WAN 2 (both directions), with the exact ping commands used and the results (replies received, 0% loss). (2 marks)
-

Written Reflection — End of Project Report (5 Marks)

The project report must include a Reflection section (300–500 words) as the last section before the Project Artifacts section. This section must be written in the students' own words and should address some of the following:

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method (e.g., static routing vs. RIP, your IP addressing scheme) over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

Version History / Commit Log Requirements (Common to Both Reports)

Students must document and submit their project history/timestamps using a live, editable link:

- **Written report:** Submit a shared Google Doc or Word Online link (sharing set to "Anyone with the link can view") along with your final PDF report. The link must show the progress of the students' work with timestamps/screenshots.
- **Code/data/configuration files:** Submit Packet Tracer files (.pkt) and any configuration scripts via a GitHub or GitLab repository link (the coordinator may choose a repository suitable for the course). The repository must show a commit history reflecting incremental progress. Commit regularly (at minimum once per week), with clear, descriptive commit messages.

A separate section in the report must be added and named "Project Artifacts" to include all links, for example:

- Live report document (with edit history): [link]
- GitHub/GitLab repository (with commit history): [link]
- Packet Tracer project file (.pkt): [link or attached]

What NOT to do: Students must not write their report elsewhere and paste the final version in at the last minute. They also must not make a single commit containing the entire project. Both are treated as red flags regardless of the quality of the final work.

Important Notes

- **AI use:** Students may use AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) only for brainstorming approaches, debugging small errors, or proofreading text. AI must not be used to generate analysis, write configuration logic, or produce substantial portions of the report or project files. Any permitted AI use must be disclosed in the Reflection section. Undisclosed AI use, or AI use beyond this scope (disclosed or not), will result in a zero grade for the entire project.
- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit the mid progress report as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.
- The mid progress report should be concise including screenshots.
- The same shared document link and repository submitted in Phase 1 must continue to be used for Phase 2, so that the full project history is visible in one place.

Work completed this period:

During the first phase, we prepared the project environment and created a GitHub repository to store the Packet Tracer files, screenshots, and project documents. A README file was created containing the group members' names, student IDs, and GitHub usernames. We also decided to use the original project Word template as our shared report document to maintain the required format and record the group's progress through the document history.

We installed Cisco Packet Tracer and became familiar with its interface and basic networking tools. We learned how to add and configure network devices such as PCs, laptops, switches, and routers and organize them into a network topology.

We also studied the different types of network connections available in Packet Tracer. We practiced using copper straight-through cables to connect end devices to switches and switches to routers. In addition, we learned the purposes of copper cross-over cables and Serial DCE/DTE cables.

We built the two required WAN topologies separately, with each WAN containing two LANs. WAN 1 includes LAN 1 (Computer Lab) and LAN 2 (Admin Office), while WAN 2 includes LAN 3 (Library) and LAN 4 (Faculty Office). We assigned static IP addresses, subnet masks, and default gateways to the end devices. We also configured the router interfaces with IP addresses that serve as the default gateways for their connected LANs.

Finally, we tested network communication using the ping command. The tests confirmed successful communication within each LAN and between the two LANs in each WAN.

Objectives and Advantages of Using Cisco Packet Tracer:

The main objective of using Cisco Packet Tracer in this project is to design, configure, and test the required network virtually. It allows us to build two WANs, each containing two LANs, connect the different network devices, assign IP addresses, configure routers, and test whether data packets can be transmitted successfully between devices in different LANs.

Packet Tracer provides several advantages for this project. It allows us to practice network configuration without using physical equipment, which reduces cost and saves time. It also provides a safe environment to test different configurations, identify and correct mistakes, and verify connectivity using the ping command. This helps us understand how end devices, switches, routers, network links, and IP addressing work together.

One key decision made during this period:

One key decision made during this period was to build and test WAN 1 and WAN 2 separately before combining them. This allowed the group members to work on different parts of the project at the same time. It also made it easier to identify and correct addressing or connectivity problems. After each WAN was successfully tested using the ping command, we planned to combine them and configure the connection between the two routers during the next phase.

One problem encountered:

One problem encountered was that two group members initially did not have GitHub accounts. This delayed adding them as collaborators and sharing the project files through the repository. The problem was resolved after both members created accounts, shared their usernames, joined the repository, and became familiar with the basic GitHub workflow.

Plan for the next period:

During the next period, we plan to combine the completed WAN 1 and WAN 2 topologies into one Packet Tracer file. We will connect Router 1 and Router 2 using a Serial DCE/DTE link and configure the serial interfaces with the required IP addresses. We will then configure static routes to enable communication between the two WANs. Finally, we will test communication between devices in all four LANs using the ping command, resolve any problems found, and document the final results for Phase 2.

Each group member's input and participation:

Abdulrahman Bin Omran:

Week 1:

Discussed the project with the group and the distribution of the project tasks. Created the GitHub repository, organized the initial folders, created and edited the README file, added the students' names, IDs, and GitHub usernames, and shared the project report document with the group.

Week 2:

Downloaded Cisco Packet Tracer and became familiar with the software.

Week 3:

Created the initial LAN 1 and LAN 3 topologies, tested connectivity between devices within each LAN using the ping command and coordinated the workflow among the group members.

Week 4:

Reviewed the combined topology and made sure it was correctly configured. Also reviewed all 16 LAN connections, confirmed that each link was active, verified that Copper Straight-Through cables were used between end devices and switches and between switches and routers, and documented the connections in the report.

Week 5:

Reviewed the final Packet Tracer file and Phase 2 report to make sure all project requirements were completed. Fixed startup configuration saving on both routers. Checked the work for Questions 1–5, reviewed the screenshots and commands, and verified that the final network configuration matched what was documented in the report. Also worked on the Reflection section and performed a final review of the report and project files before submission.

Abdullah Ali Almekrami:

Week 1:

Planned the network topology with the team and discussed the IP addressing scheme for each LAN. Participated in reviewing the project requirements and helped organize the task distribution among the group members to ensure the work was completed efficiently.

Week 2:

Installed Cisco Packet Tracer, created a GitHub account, and prepared the required tools for the project.

Week 3:

Completed the WAN 1 network configuration and verified connectivity using ping tests.

Week 4:

Completed the final network topology by creating and naming all required network devices, including the routers, switches, PCs, and laptops. Verified that the topology contained 12 end devices, 4 switches, and 2 routers, and documented the device information and naming steps for the project report.

Week 5:

I completed the WAN connection between Main-R01 and Branch-R01 using a serial link. I configured the serial interfaces by assigning IP addresses to both routers and setting the clock rate on the DCE interface. I then enabled RIP version 2 and added the required network statements so the routers could exchange routing information. Finally, I verified that the WAN connection and routing were working correctly and captured screenshots of the serial link and the router configurations for the project report.

Ali Zaid Almogherah:

Week 1:

During the first week, our team reviewed the project requirements, discussed the assignment guidelines, and outlined the overall network design. We also assigned tasks to each member according to their role to keep the project well organized and on schedule.

Additionally, I created a GitHub account, shared my username with the team, joined the project repository, and obtained access to the shared project report.

Week 2:

Downloaded Cisco Packet Tracer, verified that it was installed and working correctly, and became familiar with the program.

Week 3:

Completed the WAN 2 (LAN 3 and LAN 4) configuration and verified network connectivity using ping tests.

Week 4:

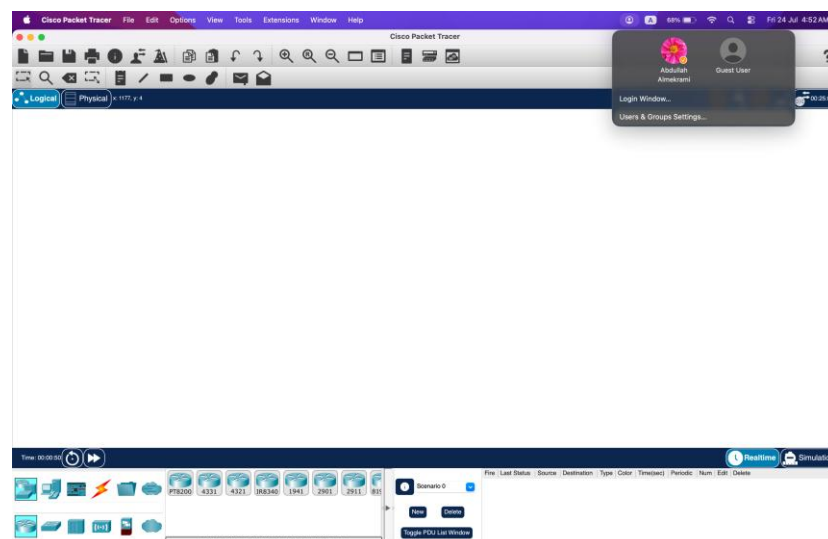
Worked on Question 3 (Assigning IP addresses). Took screenshots of the IP address, subnet mask, and default gateway on one device in each of the four LANs and verified the LAN interfaces on both routers using the CLI. Added the screenshots and the commands used to the report and uploaded the files to the GitHub repository.

Week 5:

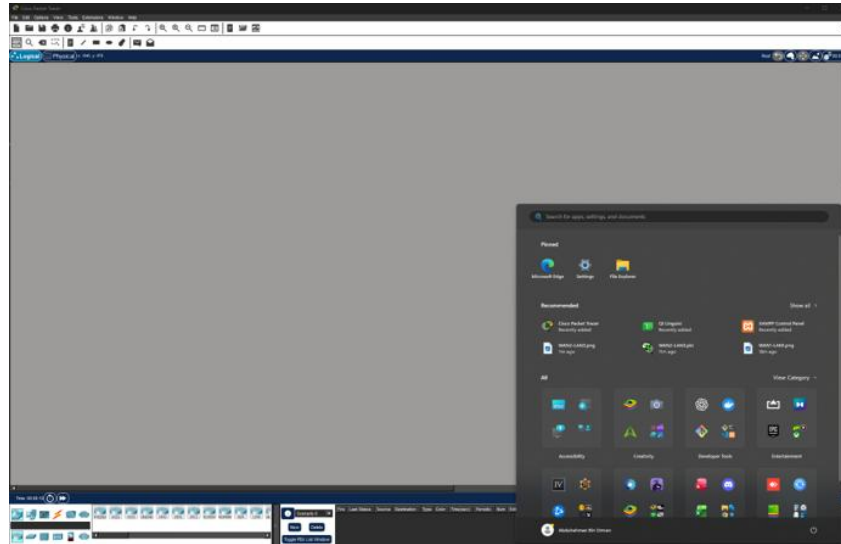
Worked on Question 5 (Testing data transmission). Ran ping tests between a device in WAN 1 and a device in WAN 2 in both directions, and both tests returned all replies with 0% loss. Added the screenshots and commands to the report and uploaded the files to the GitHub repository.

Screenshot of the Installation of Packet Tracer:

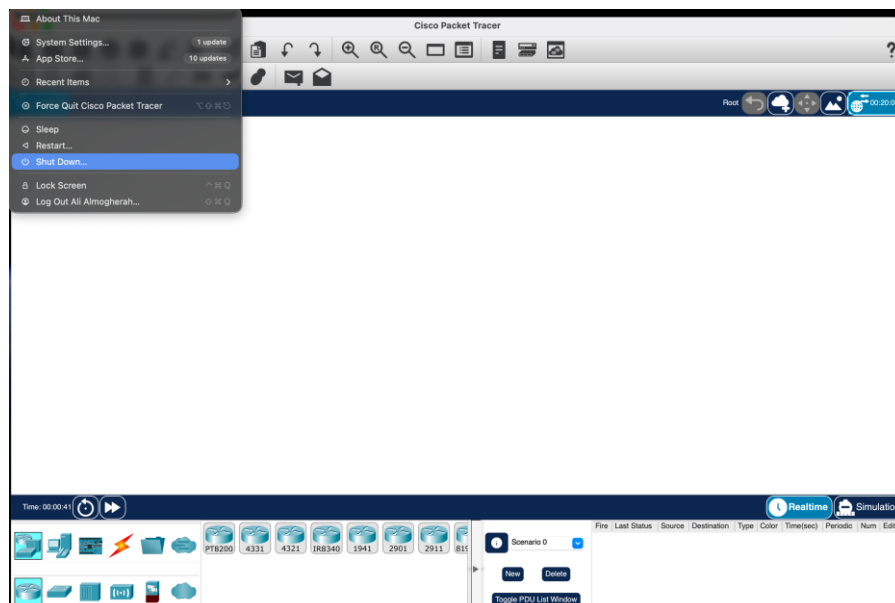
Abdullah's:



Abdulrahman's:



Ali's:

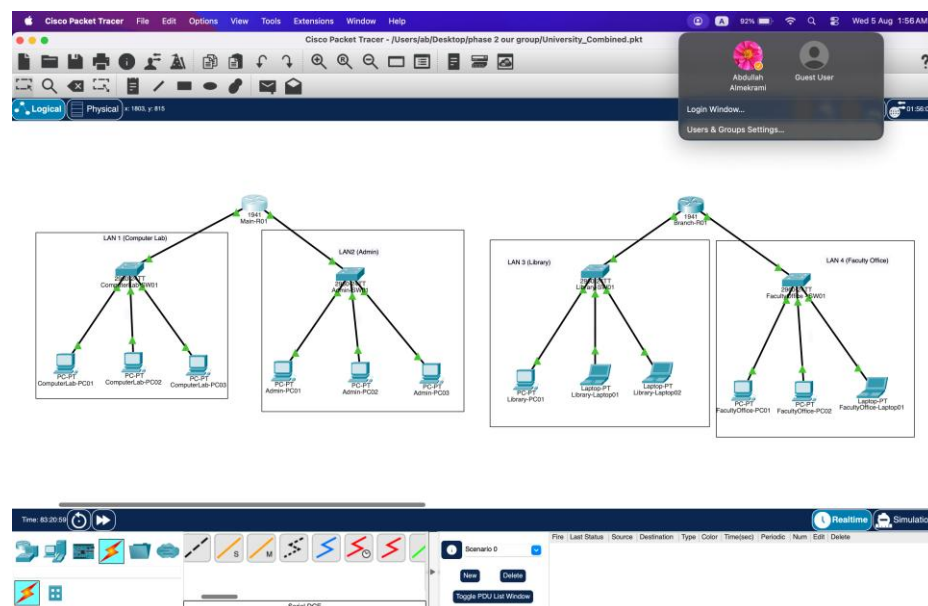


Phase 2:

1- Creating nodes:

#	Device Type	Model	Quantity	Device Name	Rename Steps
1	Router	Cisco 1941	2	Main-R01, Branch-R01	CLI: enable - configure terminal - hostname <router-name>
2	Switch	Cisco 2960-24TT	4	ComputerLab-SW01, Admin-SW01, Library-SW01, FacultyOffice-SW01	Config → Global → Display Name
3	End Devices	PC-PT, Laptop-PT	12	ComputerLab-PC01, ComputerLab-PC02, ComputerLab-PC03, Admin-PC01, Admin-PC02, Admin-PC03, Library-PC01, Library-Laptop01, Library-Laptop02, FacultyOffice-PC01, FacultyOffice-PC02, FacultyOffice-Laptop01	Config → Settings → Display Name

The network is divided into two WANs: WAN 1 and WAN 2. WAN 1 contains LAN 1 (Computer Lab) and LAN 2 (Admin Office), while WAN 2 contains LAN 3 (Library) and LAN 4 (Faculty Office). Each LAN includes three end devices connected through one switch. Overall, the network consists of 12 end devices, 4 switches, and 2 routers. This design provides a well-organized network structure and allows communication between all connected devices.



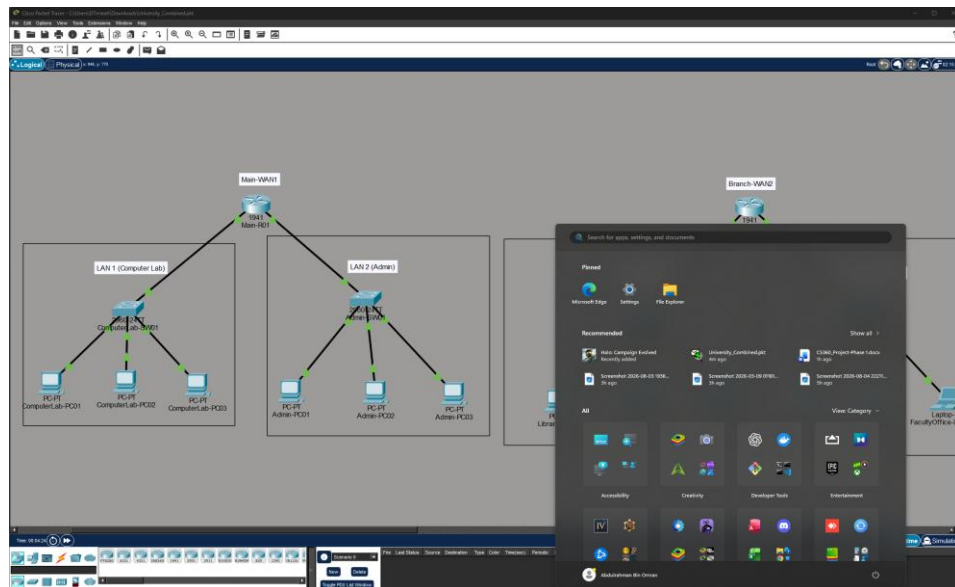
2- Giving connections in the LAN:

#	Connection Type	Quantity	Cable used	Reason
1	End device to switch	12	Copper Straight-Through	Connects different device types
2	Switch to router	4	Copper Straight-Through	Connects different device types

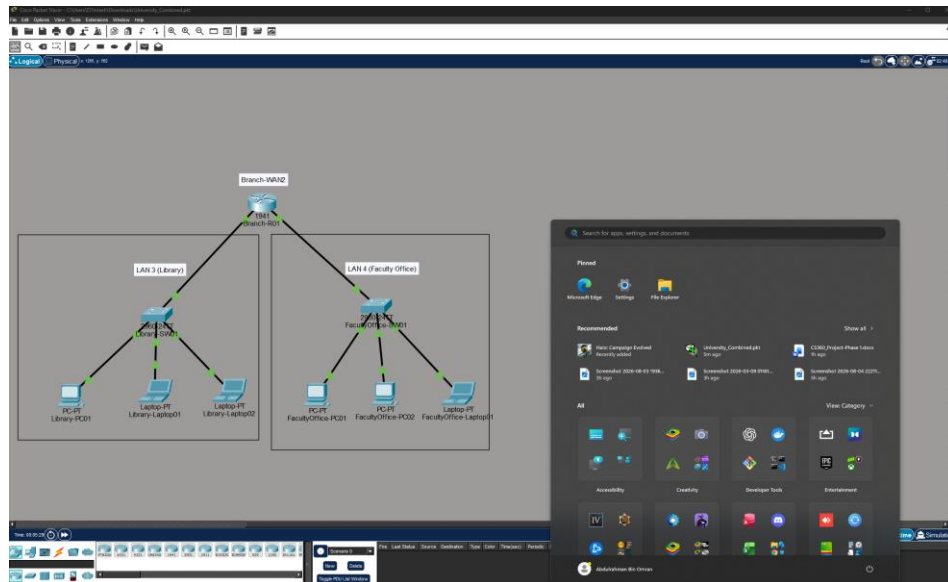
The configuration steps used:

- Selected Copper Straight-Through from Connections.
- Connected each end device's FastEthernet0 port to an available switch port.
- Connected each switch to the appropriate router interface.
- Confirmed that the link indicators turned green.

Main-WAN1 screenshot:



Branch-WAN2 screenshot:



3- Assigning IP addresses:

We assigned the IP addresses for the end devices and router LAN interfaces. Each LAN has its own network and its default gateway is the router interface connected to it:

LAN	Network	Subnet mask	Default gateway
LAN 1 (computer lab)	172.16.10.0/24	255.255.255.0	172.16.10.1
LAN 2 (Admin)	172.16.20.0/24	255.255.255.0	172.16.20.1
LAN 3 (Library)	172.16.30.0/24	255.255.255.0	172.16.30.1
LAN 4 (Faculty Office)	172.16.40.0/24	255.255.255.0	172.16.40.1

On each end device, we opened Desktop > IP Configuration, selected Static, and entered the IP address, subnet mask, and default gateway. This step was completed

through the GUI, so no CLI commands were used. The screenshots below show one device from each LAN.

IP configuration of a device in LAN 1

The screenshot shows the 'IP Configuration' tab in a network management GUI. The interface is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 Address is 172.16.10.11, Subnet Mask is 255.255.255.0, Default Gateway is 172.16.10.1, and DNS Server is 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The IPv6 Address is empty, Link Local Address is FE80::230:A3FF:FE9C:D717, Default Gateway is empty, and DNS Server is empty. The '802.1X' section has 'Use 802.1X Security' unchecked, 'Authentication' set to MD5, and empty fields for Username and Password.

Field	Value
Interface	FastEthernet0
IP Configuration	
DHCP	☐
Static	☒
IPv4 Address	172.16.10.11
Subnet Mask	255.255.255.0
Default Gateway	172.16.10.1
DNS Server	0.0.0.0
IPv6 Configuration	
Automatic	☐
Static	☒
IPv6 Address	
Link Local Address	FE80::230:A3FF:FE9C:D717
Default Gateway	
DNS Server	
802.1X	
Use 802.1X Security	☐
Authentication	MD5
Username	
Password	

IP configuration of a device in LAN 2

The screenshot shows the 'IP Configuration' tab in a network management GUI. The interface is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 Address is 172.16.20.11, Subnet Mask is 255.255.255.0, Default Gateway is 172.16.20.1, and DNS Server is 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The IPv6 Address is empty, Link Local Address is FE80::260:47FF:FE59:2B74, Default Gateway is empty, and DNS Server is empty. The '802.1X' section has 'Use 802.1X Security' unchecked, 'Authentication' set to MD5, and empty fields for Username and Password.

Field	Value
Interface	FastEthernet0
IP Configuration	
DHCP	☐
Static	☒
IPv4 Address	172.16.20.11
Subnet Mask	255.255.255.0
Default Gateway	172.16.20.1
DNS Server	0.0.0.0
IPv6 Configuration	
Automatic	☐
Static	☒
IPv6 Address	
Link Local Address	FE80::260:47FF:FE59:2B74
Default Gateway	
DNS Server	
802.1X	
Use 802.1X Security	☐
Authentication	MD5
Username	
Password	

IP configuration of a device in LAN 3

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address172.16.30.11

Subnet Mask255.255.255.0

Default Gateway172.16.30.1

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::2D0:97FF:FE03:DC96

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

IP configuration of a device in LAN 4

Physical

Config

Desktop

Programming

Attributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

DHCP

Static

IPv4 Address172.16.40.11

Subnet Mask255.255.255.0

Default Gateway172.16.40.1

DNS Server0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

Link Local AddressFE80::20A:F3FF:FE0C:6378

Default Gateway

DNS Server

802.1X

Use 802.1X Security

AuthenticationMD5

Username

Password

On the routers, we configured each LAN interface from the CLI tab using the IP address command and enabled it with no shutdown because router interfaces are shut down by default.

The screenshots below show the commands, and the show ip interface brief output confirms that all interfaces are up with the correct IP addresses.

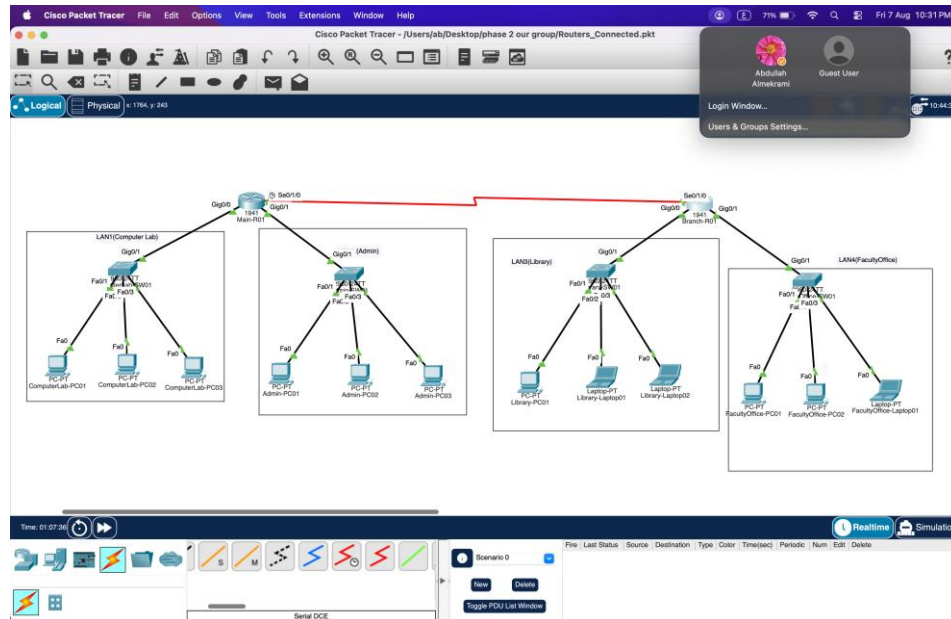
Router 1 (Main-R01): LAN interface configuration and verification

```
Main-R01>enable
Main-R01#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  172.16.10.1    YES manual up          up
GigabitEthernet0/1  172.16.20.1    YES manual up          up
Vlan1           unassigned      YES unset  administratively down down
Main-R01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Main-R01(config)#interface GigabitEthernet0/0
Main-R01(config-if)#ip address 172.16.10.1 255.255.255.0
Main-R01(config-if)#no shutdown
Main-R01(config-if)#exit
Main-R01(config)#interface GigabitEthernet0/1
Main-R01(config-if)#ip address 172.16.20.1 255.255.255.0
Main-R01(config-if)#no shutdown
Main-R01(config-if)#end
Main-R01#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 2 (Branch-R01): LAN interface configuration and verification

```
Branch-R01>enable
Branch-R01#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  172.16.30.1    YES manual up          up
GigabitEthernet0/1  172.16.40.1    YES manual up          up
Vlan1           unassigned      YES unset  administratively down down
Branch-R01#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Branch-R01(config)#interface GigabitEthernet0/0
Branch-R01(config-if)#ip address 172.16.30.1 255.255.255.0
Branch-R01(config-if)#no shutdown
Branch-R01(config-if)#exit
Branch-R01(config)#interface GigabitEthernet0/1
Branch-R01(config-if)#ip address 172.16.40.1 255.255.255.0
Branch-R01(config-if)#no shutdown
Branch-R01(config-if)#end
Branch-R01#
%SYS-5-CONFIG_I: Configured from console by console
```

4. Creating the WAN connection:



Main-R01 Serial Interface Configuration

```
Main-R01(config)#interface Serial0/0/0  
Main-R01(config-if)#ip address 10.10.10.1 255.255.255.252  
Main-R01(config-if)#clock rate 2000000  
Main-R01(config-if)#no shutdown
```

Branch-R01 Serial Interface Configuration

```
Branch-R01(config)#interface Serial0/0/0  
Branch-R01(config-if)#ip address 10.10.10.2 255.255.255.252  
Branch-R01(config-if)#no shutdown
```

RIP Configuration, Main-R01

```
Main-R01(config)#router rip
Main-R01(config-router)#version 2
Main-R01(config-router)#no auto-summary
Main-R01(config-router)#network 10.0.0.0
Main-R01(config-router)#network 172.16.0.0
```

RIP Configuration, Branch-R01

```
Branch-R01(config)#router rip
Branch-R01(config-router)#version 2
Branch-R01(config-router)#no auto-summary
Branch-R01(config-router)#network 10.0.0.0
Branch-R01(config-router)#network 172.16.0.0
```

5. Successful data transmission using the ping command:

To test the connection between the two WANs, we used the ping command between a device in WAN 1 and a device in WAN 2, in both directions.

First, from the device 172.16.10.11 in LAN 1 (Main Campus), we pinged the device 172.16.30.11 in LAN 3 (Branch Campus):

```
C:\>ping 172.16.30.11

Pinging 172.16.30.11 with 32 bytes of data:

Reply from 172.16.30.11: bytes=32 time=2ms TTL=126
Reply from 172.16.30.11: bytes=32 time=1ms TTL=126
Reply from 172.16.30.11: bytes=32 time=3ms TTL=126
Reply from 172.16.30.11: bytes=32 time=3ms TTL=126

Ping statistics for 172.16.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\>
```

From the device 172.16.30.11 in LAN 3, we pinged back the device 172.16.10.11 in LAN 1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.10.11

Pinging 172.16.10.11 with 32 bytes of data:

Reply from 172.16.10.11: bytes=32 time=2ms TTL=126
Reply from 172.16.10.11: bytes=32 time=3ms TTL=126
Reply from 172.16.10.11: bytes=32 time=2ms TTL=126
Reply from 172.16.10.11: bytes=32 time=3ms TTL=126

Ping statistics for 172.16.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>|
```

In both tests, all 4 packets were received with 0% loss, which shows that data is transmitted successfully between the two WANs.

Reflection:

Routing Method:

Our initial plan was to use static routing to connect WAN 1 and WAN 2. Since there are only two routers, static routing would have been suitable for this project, as the required routes could be configured manually without much complexity. When we implemented the WAN connection in Phase 2, we used RIP version 2 instead. RIP v2 allows the routers to exchange routing information automatically instead of requiring us to manually configure a route to every network. We decided to use RIP version 2 in the final topology because it provided communication between all four LANs and made route management more automatic. Static routing would give us more direct control, while RIP version 2 automatically exchanges routing information and would make it easier to expand the network later.

IP Addressing Scheme:

For the IP addressing scheme, we used a separate /24 network for each LAN: 172.16.10.0/24, 172.16.20.0/24, 172.16.30.0/24, and 172.16.40.0/24. This made it easy to identify which LAN a device belongs to and kept the four LANs logically separated. For the serial connection between the two routers, we used the 10.10.10.0/30 network because the connection only requires two usable IP addresses, one for each router. This addressing structure also made configuring and troubleshooting the topology easier.

AI disclosure:

ChatGPT and Claude AI tools were used during the project for brainstorming approaches, proofreading the report for grammar and clarity, and helping review and troubleshoot small issues. The group reviewed the suggestions and compared them with the project requirements before using them. Suggested wording was reviewed and rewritten by the students in their own words. The Packet Tracer configurations, connectivity tests, screenshots, and final project decisions were completed and verified by the group members.

What We Would Do Differently:

If we were to redo the project, we would combine WAN 1 and WAN 2 earlier in the project. Building the two WANs separately helped us divide the work, build both WAN simulations, and test each side individually. However, building them in the same Packet Tracer file from an earlier stage would have given us more time to configure the serial connection, test routing between both WANs, reduce the need to repeat connection checks after combining the files, and solve any problems before the final stage.

Project Artifacts:

- Live report document: [CS360_Project-final.docx](#)
- GitHub/GitLab repository: [CS360_Project GitHub](#)
- Packet Tracer project file (.pkt): [Packet Tracer Files](#)
- Packet Tracer Final project file (.pkt): [Packet Tracer Final File](#)